

## KYZYLZHAR, Kazakhstan

WMO: 355760

Lat: 48.300N

Lon: 69.650E

Elev: 361

StdP: 97.06

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -31.2      | -28.5 | -34.9                         | 0.1 | -30.9 | -32.0 | 0.2 | -28.3 | 12.5                  | -7.7 | 10.4 | -6.7 | 2.0                   | 50   | 0.570 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.5                   | 35.0            | 18.6 | 33.3 | 17.8 | 31.6 | 17.4 | 19.7                | 30.8 | 18.9 | 30.2 | 18.2 | 29.2 | 3.2                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.3                            | 12.1 | 22.6 | 15.2 | 11.3 | 22.9 | 13.9 | 10.4 | 23.0 | 58.1          | 30.7 | 55.3 | 30.2 | 53.1 | 29.0 | 23.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.1          | 8.6  | 7.4 | -33.2                      | 37.9 | 3.3                | 1.5 | -35.5                                              | 39.0 | -37.5      | 39.9 | -39.3      | 40.8 | -41.7      | 41.9 |    |    |
| (5)               |      |     | -33.3                      | 21.0 | 3.2                | 1.3 | -35.6                                              | 22.0 | -37.5      | 22.7 | -39.4      | 23.5 | -41.7      | 24.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 5.5    | -15.5 | -14.2 | -3.5  | 9.2   | 15.7  | 21.3  | 23.0  | 21.5  | 14.8  | 6.4   | -2.9  | -11.2 |  |  |
|                                                                        | DBStd       | 14.88  | 7.13  | 7.36  | 7.34  | 5.12  | 4.24  | 3.79  | 3.31  | 3.78  | 4.67  | 4.99  | 6.66  | 7.41  |  |  |
|                                                                        | HDD10.0     | 3156   | 791   | 679   | 422   | 73    | 6     | 0     | 0     | 0     | 9     | 131   | 387   | 658   |  |  |
|                                                                        | HDD18.3     | 5102   | 1049  | 912   | 678   | 275   | 103   | 16    | 4     | 14    | 127   | 370   | 637   | 917   |  |  |
|                                                                        | CDD10.0     | 1502   | 0     | 0     | 2     | 49    | 182   | 339   | 402   | 357   | 152   | 19    | 1     | 0     |  |  |
|                                                                        | CDD18.3     | 406    | 0     | 0     | 0     | 1     | 21    | 104   | 147   | 112   | 21    | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 5412   | 0     | 0     | 1     | 35    | 381   | 1415  | 1853  | 1408  | 312   | 7     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 2326   | 0     | 0     | 0     | 6     | 99    | 624   | 863   | 616   | 117   | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.0    | 2.9   | 3.1   | 3.5   | 3.4   | 3.3   | 3.1   | 2.9   | 2.8   | 2.6   | 2.7   | 2.8   | 3.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 193    | 17    | 12    | 13    | 17    | 22    | 16    | 21    | 13    | 10    | 16    | 15    | 16    |  |  |
|                                                                        | PrecMax     | 325    | 70    | 33    | 39    | 49    | 69    | 80    | 83    | 54    | 37    | 51    | 50    | 37    |  |  |
|                                                                        | PrecMin     | 87     | 3     | 1     | 1     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 4     | 4     |  |  |
|                                                                        | PrecStd     | 52     | 13    | 7     | 10    | 14    | 19    | 16    | 19    | 12    | 9     | 13    | 10    | 9     |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 1.0   | 2.0   | 17.5  | 27.1  | 31.7  | 36.4  | 37.4  | 36.3  | 33.9  | 24.2  | 13.9  | 2.4   |  |  |
|                                                                        |             | MCWB   | -0.2  | 0.3   | 9.2   | 14.3  | 17.0  | 19.4  | 19.8  | 18.9  | 17.6  | 12.5  | 9.7   | 0.3   |  |  |
|                                                                        | 2%          | DB     | -0.4  | 0.5   | 12.7  | 23.7  | 29.3  | 33.9  | 35.3  | 34.1  | 30.2  | 20.4  | 10.3  | 1.0   |  |  |
|                                                                        |             | MCWB   | -1.5  | -0.5  | 7.5   | 12.7  | 15.8  | 18.0  | 18.8  | 17.9  | 16.0  | 10.7  | 7.7   | -0.3  |  |  |
|                                                                        | 5%          | DB     | -2.4  | -0.9  | 8.5   | 21.0  | 27.4  | 32.2  | 33.5  | 32.2  | 27.0  | 17.5  | 7.4   | -0.1  |  |  |
|                                                                        |             | MCWB   | -3.4  | -1.9  | 5.2   | 11.8  | 15.0  | 17.5  | 18.2  | 17.3  | 14.6  | 9.4   | 4.7   | -1.3  |  |  |
|                                                                        | 10%         | DB     | -4.6  | -2.6  | 4.9   | 18.3  | 25.5  | 30.4  | 31.4  | 30.3  | 24.1  | 14.5  | 4.9   | -1.6  |  |  |
|                                                                        |             | MCWB   | -5.5  | -3.5  | 2.6   | 10.5  | 14.2  | 17.0  | 17.8  | 16.9  | 13.3  | 8.1   | 2.8   | -2.7  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | -0.2  | 0.6   | 10.2  | 15.6  | 18.8  | 20.7  | 21.3  | 19.6  | 17.9  | 13.1  | 10.7  | 0.8   |  |  |
|                                                                        |             | MCDB   | 0.7   | 1.4   | 16.3  | 23.4  | 30.2  | 32.8  | 33.4  | 31.4  | 31.3  | 21.8  | 12.9  | 1.7   |  |  |
|                                                                        | 2%          | WB     | -1.4  | -0.4  | 7.6   | 13.8  | 16.8  | 19.3  | 20.1  | 18.9  | 16.5  | 11.2  | 7.8   | -0.3  |  |  |
|                                                                        |             | MCDB   | -0.4  | 0.5   | 12.3  | 22.2  | 27.0  | 30.4  | 30.8  | 31.6  | 29.5  | 18.5  | 9.9   | 0.8   |  |  |
|                                                                        | 5%          | WB     | -3.3  | -1.9  | 5.2   | 12.5  | 15.7  | 18.4  | 19.2  | 18.1  | 14.9  | 9.9   | 5.0   | -1.3  |  |  |
|                                                                        |             | MCDB   | -2.3  | -0.9  | 8.5   | 19.6  | 25.3  | 29.5  | 29.9  | 30.0  | 25.9  | 16.3  | 6.8   | -0.1  |  |  |
|                                                                        | 10%         | WB     | -5.5  | -3.5  | 2.8   | 11.1  | 14.8  | 17.6  | 18.5  | 17.3  | 13.7  | 8.7   | 2.8   | -2.7  |  |  |
|                                                                        |             | MCDB   | -4.6  | -2.6  | 4.9   | 17.4  | 23.7  | 28.6  | 29.0  | 28.7  | 23.3  | 14.2  | 4.6   | -1.6  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.6   | 9.9   | 9.3   | 12.3  | 14.4  | 13.9  | 13.5  | 13.7  | 13.5  | 11.7  | 8.2   | 8.1   |  |  |
|                                                                        |             | MCDBR  | 6.2   | 7.6   | 12.7  | 16.7  | 17.3  | 16.5  | 16.4  | 16.2  | 17.1  | 16.4  | 10.4  | 7.1   |  |  |
|                                                                        |             | MCWBR  | 5.8   | 7.2   | 8.3   | 8.1   | 7.4   | 5.9   | 5.5   | 6.0   | 7.3   | 9.1   | 7.6   | 6.2   |  |  |
|                                                                        | 5% WB       | MCDBR  | 6.3   | 7.8   | 12.2  | 14.4  | 15.3  | 14.6  | 14.4  | 15.1  | 16.1  | 14.6  | 9.4   | 6.9   |  |  |
|                                                                        |             | MCWBR  | 6.0   | 7.5   | 8.2   | 8.0   | 7.2   | 6.0   | 5.7   | 6.3   | 7.4   | 8.8   | 7.3   | 6.2   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.305 | 0.340 | 0.390 | 0.419 | 0.428 | 0.422 | 0.429 | 0.395 | 0.352 | 0.347 | 0.336 | 0.311 |  |  |
|                                                                        | taud        |        | 2.066 | 1.985 | 2.042 | 2.092 | 2.124 | 2.212 | 2.207 | 2.284 | 2.357 | 2.279 | 2.183 | 2.119 |  |  |
|                                                                        | Ebn at Noon |        | 722   | 779   | 807   | 826   | 835   | 842   | 829   | 844   | 850   | 780   | 686   | 661   |  |  |
|                                                                        | Edh at Noon |        | 91    | 125   | 140   | 148   | 149   | 138   | 136   | 121   | 101   | 93    | 82    | 77    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.63  | 2.82  | 4.00  | 5.18  | 6.58  | 6.97  | 6.56  | 5.97  | 4.61  | 2.71  | 1.40  | 1.14  |  |  |
|                                                                        | RadStd      |        | 0.25  | 0.21  | 0.38  | 0.47  | 0.42  | 0.40  | 0.34  | 0.39  | 0.23  | 0.27  | 0.14  | 0.17  |  |  |

## Historical Trends

|      |                        | DBAvg      | Heating    |            | Cooling    |            |            | Degree-Days |            |            |            |      |
|------|------------------------|------------|------------|------------|------------|------------|------------|-------------|------------|------------|------------|------|
|      |                        |            | 99% DB     | 99% DP     | 1% DB      | 1% WB      | 1% DP      | HDD10.0     | HDD18.3    | CDD10.0    | CDD18.3    |      |
| (46) | Station Only           | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A  | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (46) |
| (47) | Regional (3 neighbors) | N/A        | N/A        | N/A        | N/A        | N/A        | N/A        | N/A         | N/A        | N/A        | N/A        | (47) |

Nomenclature: See separate page